Department of Electrical and Computer Systems Engineering
ECE2071 - Systems Programming

ECE2071 Project Milestone

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Revision History

Use this revision history to monitor for any changes we may need to make to this document to clarify requirements or provide hints

Description	Version	Date
Initial release	v1.0	14/03/2026

Introduction

This task is designed to be completed over the first several weeks of the ECE2071 project, supported by material introduced in the Week 4 and Week 5 lab sessions. It will be presented to the unit coordinator in Week 6. The milestone as a whole is worth 5% towards your final grade, and is structured as a base set of requirements that equate to a pass grade (50%), with each subsequent grade band requiring additional functionality to be implemented.

The primary goal of this task is to assist you and your team in meeting the communications requirement for the project, as we found many students lacking in this area in previous cohorts. By completing the UART requirement within this milestone, you will hopefully have the skills necessary to tackle the main project.

The **learning outcomes** for this milestone are as follows:

- Use documentation and IDE tools to setup and control a microcontroller's peripherals.
- Construct and demonstrate a sequence of UART communications between multiple devices.

Academic Integrity and AI

Academic Integrity: You may discuss **general** details (eg: approaches) of the milestone, and other project matters, with other students outside your team. However, **you may not:**

- write the code together with students from other teams,
- show students from other teams your code,
- copy another team's work, or
- present work you have found online as your own (i.e. without attribution),

This is plagiarism and/or collusion and is prohibited by Monash's academic integrity policy. Breaches of this policy attract serious consequences.

If you research some concept online or re-use code from your labs (or from this unit) you must cite it (any citation method is acceptable).

AI Usage: In this assessment, you may use generative AI in order to develop your understanding of your code (eg "explain how this works to me", or "help me find the bug in this code") only.

You may not use it to directly generate programs for submission. Any use of generative AI for this purpose must be explicitly acknowledged in a comment, explaining your prompts and how you used the outputs. If you do not acknowledge your use of AI, it may be submitted as plagiarism under the academic integrity guidelines. Any work submitted for a mark must:

- represent a sincere demonstration of your human efforts, skills and subject knowledge that you will be accountable for.
- adhere to the guidelines for AI use set for the assessment task.
- reflect the University's commitment to academic integrity and ethical behaviour.

- Inappropriate AI use and/or AI use without acknowledgement will be considered a breach of academic integrity.

Demonstration

Students will present their milestones as a team during the allocated **Week 6 lab session**.

Special Consideration: Short extensions are not available due to the nature of this task as an in-person, group based milestone. Special consideration is available, but will only apply to the individual student it is granted to, with either a rescheduled demonstration or an alternative milestone/project assigned.

Project Milestone Task

The base milestone functionality represents a minimal recreation of a simple (now obsolete) network topology known as a 'token ring network'. The key principle behind token ring networks is that a 'token' is passed around a circle of devices. The device in possession is the only one allowed to transmit data. Once it has finished transferring, the device will pass the token onto the next in the chain.

For this activity, you must create a token ring network by daisy-chaining the STM32s' TX and RX pins, allowing a packet (token) to be received from the previous device and transmitted to the next. In addition, one of the STMs in the chain will be designated the 'head' device, which will be responsible for communicating with a Python script.

When we need to refer to the 'ID' of an STM, we will use the following format: 2-digit <Team_ID> for your team (eg 01 for Team 1, 22 for Team 22, etc), and underscore, and then a unique integer representing the STMs position in the ring. For example, Team 22's STMs would be 22_0, 22_1, 22_2, 22_3 and 22_4.

There are three levels of expanding requirements you can complete for this milestone, corresponding to 3 possible grades. Stream requires all of Basic, Full requires all of Stream.

- Basic (50% - 2.5%/5% - Pass),
- Stream (75% - 3.5%/5% - Distinction), or
- Full (100% - 5%/5% - High Distinction)

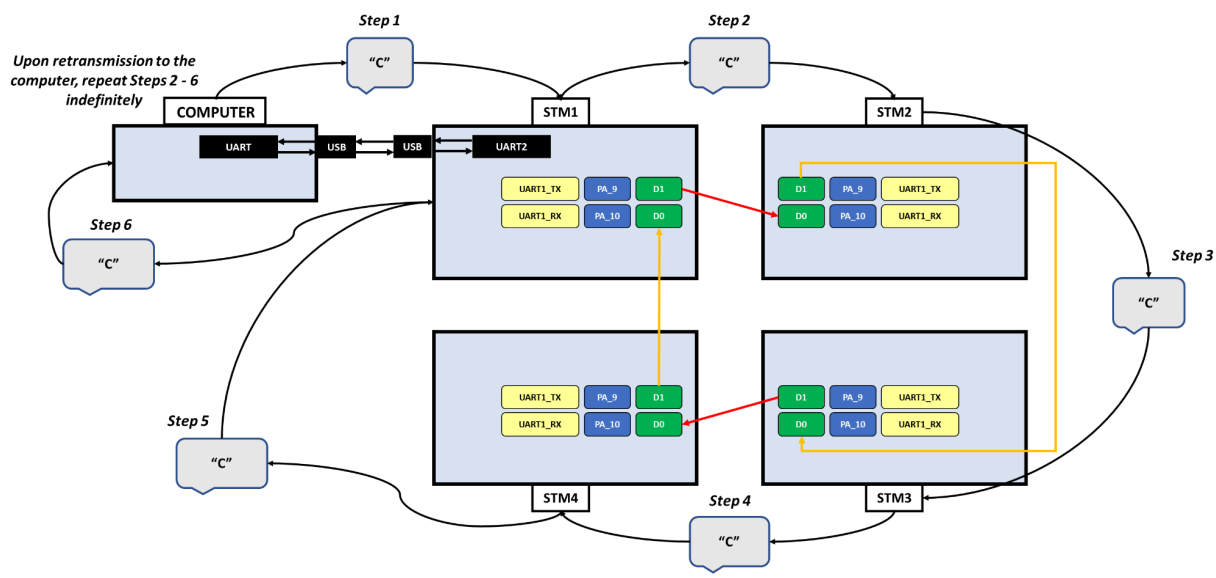
Individual marks for this milestone will be set via a short viva conducted during/after the demonstration in Week 6.

Basic Requirements:

To receive a **passing grade (2.5%)** for this milestone, your system must meet the following requirements:

- Your token ring network must contain **at least** N - 1 STM32s (where N is the number of members in your team, adjusted for any team members who are absent[1]).
- Your token ring network must send a single byte as the 'token'. It does not matter what this byte is for the pass level.
- The 'head' STM can be any device in the chain and designated by your team.
- After receiving the token, the STM should hold it for **250ms** before passing it on to the next device in the chain.
- Each STM32 should use its onboard green LED to indicate when it has possession of the token.
- Communication should begin when the 'head' STM sends the first packet that is triggered by receiving an arbitrary message from a PC over UART.
- Whenever the 'head' STM receives the token (indicating an entire loop), it should send it over UART to the PC to be displayed on any console [e.g. Termite or a Python handler] and then pass it on to the next device.

The functionality that matches the basic requirements of this milestone is depicted in the figure below.



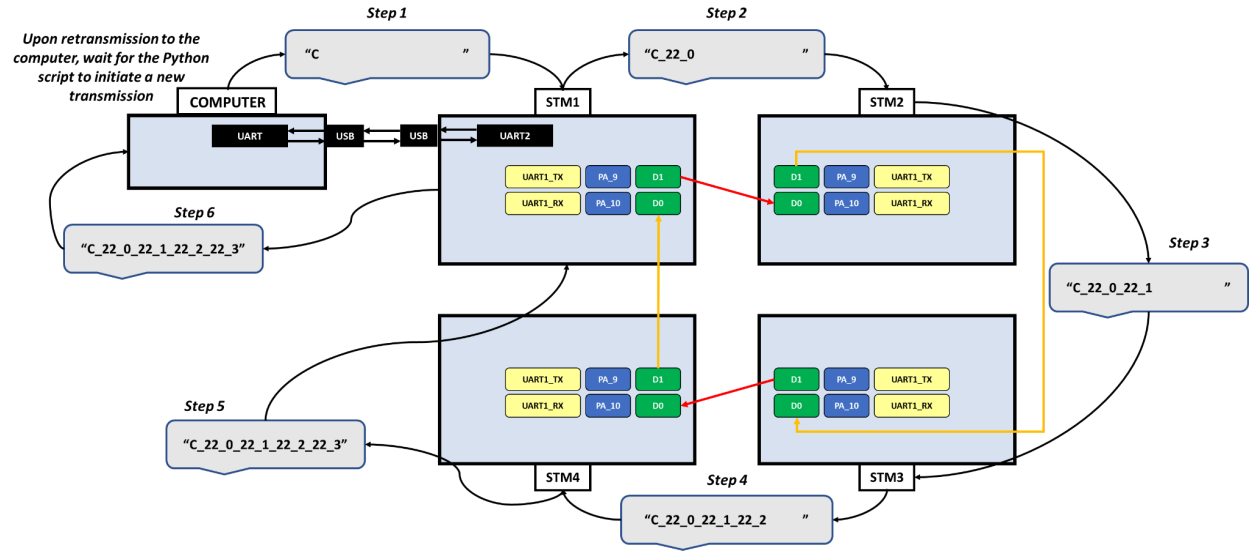
Stream Requirements:

To receive a **distinction grade (3.5%)** for this milestone, your system must **also** meet the following requirements:

- Your token ring network must contain N STM32s (where N is the number of members in your team).
- Instead of an arbitrary byte, the 'head' STM must receive a message from a Python script as the initial message.
- Each device must append its ID [defined previously] to the received string before passing it on.
- When the 'head' STM receives the final message, it will still send it to the PC to be displayed and **not** start the message over again until the Python script requests it to do so.

[1] Absent team members receive a score of 0 for the milestone unless they have spec con

The functionality that matches the stream requirements of this milestone (considering Team 22 as an example) is depicted in the figure below.



Full Requirements:

To receive a **high-distinction grade (5%)** for this milestone, your system must **also** meet the following requirements:

- A 'checksum' must be introduced to the communication system between each STM to ensure the data is not being corrupted.
    - The checksum is calculated by performing a bitwise XOR[2] between each **byte** in the received string (excluding the previous checksum). For example, if the string has 4 bytes, the final result would be ((B0 ^ B1) ^ B2) ^ B3).
    - Once calculated, the STM should verify the calculated checksum matches the received checksum. If it does not, the STM should halt the ring and leave its LED on.
    - After appending its name to the original message, the STM must now re-calculate the checksum and append it to the end of the message before sending it on.
- The 'head' STM must be dynamic (may not be STM1 as in the previous cases) and will be chosen at random by the unit coordinator during your demonstration.
    - This means every STM must be capable of acting as the 'head' **without** reprogramming

[2] A bitwise XOR of two bytes is done by using the '^' operator between two bytes in C.

Demonstration and Marking

At the beginning of your allocated week 6 lab session, you will have 2 minutes to present your token ring network to the unit coordinator. During your demonstration, you must demonstrate that your system meets **all criteria** for the mark you wish to be awarded. It is **your responsibility** to make the completion of each point as clear as possible.

After presenting, a viva-voce will be conducted for every team member to determine their understanding of the system and contribution to the milestone. Based on the level of understanding demonstrated by the team member, the grade of the project milestone will be scaled as shown in the table below.

Category Description	**Detailed Description**	**Scaling Factor**
Expert understanding	The student is in full command of every aspect of the design, and demonstrates a **deep familiarity of the design.** They can answer questions precisely, correctly, and concisely with no prompting.	100%
Complete understanding	The student is **well prepared** and can **consistently provide answers** that are **fully correct**, possibly with some prompting. The student may lack confidence or speed in answering.	90%
Tolerable understanding	The student is **reasonably prepared**, and gives answers that are **partially correct, or fully correct in one area,** but cannot answer questions about another area correctly.	80%
Selective understanding	The student has likely seen the design, and can answer **something partially relevant or correct** to a question but they **can't engage in a serious discussion of the work**.	70%
Trivial understanding	The student may have seen the design/circuits before, and can answer **something partially relevant or correct** to a question but they **clearly can't engage in any discussion of the work**.	60%
No Understanding	The student has **not prepared**, **cannot answer even the most basic questions** and likely has not even seen the design/circuits before, or did not attend the interview.	0%